

SurveyReview: A Reviewer-Aligned Benchmark for Survey Evaluators

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Abstract

The rapid advancement of large language models has transformed survey writing from a months-long manual effort into an automated process. As generation scales, reliable evaluation becomes the bottleneck, and LLMs are increasingly used as survey evaluators. However, existing approaches largely rely on off-the-shelf LLM-as-a-judge methods without systematic alignment to human reviewers, and there remains a lack of systematic frameworks for quantifying alignment with human reviewers. To address this gap, we propose SurveyReview, a reviewer-aligned, multi-dimensional benchmark and dataset for survey evaluation. We collect and annotate 675 survey papers with 1,630 review reports. We structure authentic peer-review reports by converting free-form comments into four-dimensional scores (Readability, Criticalness, Comprehensiveness, Structure) paired with supporting rationales. We further release standardized train/test splits and an evaluation protocol to measure alignment between automatic evaluators and human reviewers. To validate the benchmark, we develop SurveyAlign, a strong baseline evaluator by fine-tuning Qwen3-32B with LoRA on our annotated data, augmented with external knowledge for knowledge-intensive dimensions. On the test set, SurveyReviewer substantially improves reviewer alignment over prompt-based judging with GPT-5.2, reducing average MSE from 2.28 to 1.38 and MAE from 1.15 to 0.69 across all four dimensions. Our contributions

are twofold: (1) we establish the first multi-dimensional, reviewer-aligned dataset with a reproducible evaluation framework for survey reviewing; (2) we develop a strong baseline evaluator that substantially improves alignment with human reviewers, providing a competitive reference for future research. Our code and data are available at <https://github.com/brighterr/SurveyReview>

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Benchmark, Automatic Evaluation, Large Language Models, Natural Language Processing

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1 Introduction

Large language models (LLMs) have transformed survey paper generation from a months-long manual process into an automated pipeline that retrieves literature, structures content, and drafts comprehensive manuscripts within hours [3, 10, 25]. Systems such as DeepResearch [5, 9] orchestrate retrieval, clustering, outline generation, and multi-round refinement to produce surveys that appear fluent and comprehensive in coverage. However, this automation creates a critical evaluation bottleneck: while generation scales rapidly, quality assessment still relies on human peer review. [3] Although LLMs are increasingly used as automatic evaluators [16], existing approaches largely apply off-the-shelf *LLM-as-a-judge* prompting[30] without systematic alignment to expert reviewers, and there is no rigorous framework to quantify such alignment.

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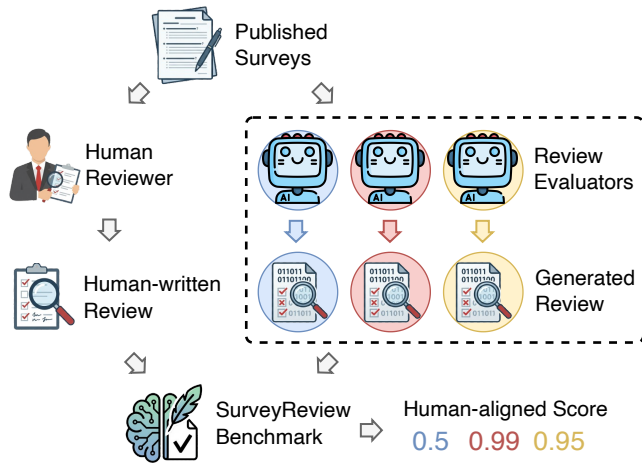


Figure 1: SurveyReview Benchmark: Evaluating to what extent the reviews produced by existing survey evaluators align with those of human expert reviewers.

An effective framework must satisfy three requirements that human reviewers naturally provide: (1) multi-dimensional assessment across stable quality dimensions (e.g., readability, structure, comprehensiveness, criticalness); (2) interpretable rationales that explain why each score was assigned, enabling targeted revision; and (3) quantifiable alignment with human judgments, measured via error metrics (MSE, MAE) to reveal systematic biases and ensure reliability. Simply producing scores is insufficient—without structured rationales and measurable alignment, automated evaluators cannot support iterative improvement or establish credibility.

Prior work evaluates surveys from reference-based perspectives rather than directly modeling expert peer review. SurveyScope [17] formalizes evaluation through reference-based comparison across curated survey datasets. SurGE [19] and DeepSurvey-Bench [28] compare generated surveys against expert-written references, scoring dimensions like coverage, citation accuracy, and “academic value” via LLM-as-a-judge. While valuable, these approaches treat expert surveys as gold-standard texts rather than learning from expert review reports. Consequently, they lack: (1) quality dimensions grounded in actual reviewing practice across diverse topics; (2) structured “score + rationale” representations that unify expert judgments into learnable signals; and (3) explicit alignment quantification via MSE/MAE to measure deviation from human reviewers. No prior work constructs evaluation datasets from peer review reports or systematically measures alignment error across quality dimensions.

We take a reviewer-aligned perspective and build an evaluation framework directly from real peer-review data. Specifically, we propose **SurveyReview**, a reviewer-aligned, multi-dimensional benchmark and dataset for survey evaluation. We collect and annotate 675 survey papers with 1,630 review reports, deriving four core dimensions consistently emphasized by experts: readability, structure, comprehensiveness, and criticalness. Readability captures clarity and ease of understanding; Structure measures the logical organization and coherence of sections; Comprehensiveness

evaluates the breadth and relevance of literature coverage; Criticalness assesses analytical depth and the extent of original insight beyond summarization. We convert free-form review comments into structured dimension-specific scores paired with supporting rationales, and release standardized splits and an evaluation protocol for reproducible alignment measurement.

Building upon this dataset, we develop **SurveyAlign**, an evaluator based on Qwen3-32B with LoRA adaptation. In addition to standard supervised fine-tuning, we incorporate reference-expansion-based Knowledge Augmentation for knowledge-intensive dimensions. This design reflects the role of references in expert review and strengthens coverage and analytical assessment.

We evaluate automatic evaluators along two aspects: score alignment and reasoning quality. Alignment with human reviewers is measured using MSE and MAE across dimensions, while rationale quality is assessed based on semantic consistency with reference explanations. We further integrate these components into a unified **Human-aligned Score (HAS)**, which jointly reflects scoring accuracy and reasoning fidelity with respect to expert peer review. Experiments show that SurveyAlign substantially improves reviewer alignment, reducing average MSE from 2.28 to 1.38 and MAE from 1.15 to 0.69 compared with GPT-5.2.

We summarize our key contributions as follows:

Dataset. We establish the first multi-dimensional, reviewer-aligned dataset derived from peer review reports. Under a unified rubric, each paper receives fine-grained scores and structured rationales across four dimensions (readability, structure, comprehensiveness, criticalness). We also provide standardized train/test splits and an evaluation protocol to measure reviewer alignment in a reproducible way.

Evaluator. We develop SurveyAlign, a strong baseline evaluator built on Qwen3-32B with LoRA. Beyond text-only fine-tuning, we incorporate a modular knowledge augmentation toolkit that enables structured integration of relevant domain knowledge for knowledge-intensive dimensions such as comprehensiveness and criticalness. On the test set, SurveyAlign reduces average MSE from 2.28 to 1.38 and MAE from 1.15 to 0.69 compared with GPT-5.2.

2 Related Work

2.1 Survey Generation

Automatic survey generation has attracted increasing attention as the volume of scientific literature continues to grow. Existing systems primarily focus on improving long-form generation pipelines through retrieval, planning, and multi-stage writing.

AutoSurvey [25] presents an end-to-end pipeline combining literature retrieval, outline construction, and section-wise generation, highlighting evaluation as a critical bottleneck. SurveyGen [3] introduces a planning-based framework with memory-guided writing to improve global consistency in long survey articles. SurveyX [10] addresses context limitations via structured knowledge organization and multi-stage generation to enhance coverage and citation accuracy. LLM×MapReduce-V2 [24] further explores test-time scaling strategies for synthesizing long-form academic content from extremely long resources.

While these approaches significantly advance generation quality, evaluation is often not the primary focus, often relying on coverage heuristics or LLM-based judgment.

2.2 Survey Evaluation

Existing work on survey evaluation can be categorized according to the primary source of evaluation signals: (1) LLM-centered scoring, (2) reference-grounded comparison, and (3) task-driven assessment.

LLM-centered scoring relies on large language models as primary evaluators to assess generated surveys according to predefined rubrics. SurGE [19] and DeepSurvey-Bench [28] adopt rubric-guided LLM scoring, sometimes complemented with auxiliary automatic metrics or reference-based statistics. While scalable, these approaches predominantly depend on model-generated judgments rather than structured supervision derived from authentic peer-review annotations.

Reference-grounded evaluation measures quality relative to expert-written surveys. SurveyScope (SciSage) [17] evaluates generated surveys against curated references in terms of content coverage and citation usage. Traditional metrics such as ROUGE [11], BLEU [13], and BERTScore [29], are also applied to long-form academic text [16].

Task-driven assessment evaluates research agents through structured academic tasks rather than direct survey comparison. DeepResearch Bench [5] and ReportBench [9] assess systems via expert-designed report-level protocols and task completion criteria.

Despite their scalability, these paradigms primarily derive supervision from model judgments, reference texts, or task outcome signals, rather than from real-world peer-review annotations. They do not explicitly model authentic expert peer-review behavior, nor do they provide dimension-level reviewer scores paired with rationales for alignment-based measurement. In contrast, SurveyReview derives structured evaluation signals from authentic peer-review reports and quantifies reviewer-aligned error across dimension-level scores paired with real reviewer rationales.

3 Task Formulation

We formalize survey evaluation as a multi-dimensional scoring task that mimics expert peer review. Given a survey paper $\mathcal{P}$, our goal is to produce a structured evaluation $\mathcal{E}$ aligning with expert judgments:

$$\mathcal{E} = f(\mathcal{P}) = \{(d_i, s_i, r_i)\}_{i=1}^n \quad (1)$$

where $d_i \in \mathcal{D}$ is a quality dimension, s_i is the numeric score, r_i is the supporting rationale, and n is the number of dimensions.

Evaluation protocol. We measure evaluator alignment with human reviewers using annotated surveys as ground truth. For each survey paper $\mathcal{P}$, the corresponding expert evaluation is denoted as $\mathcal{E}^* = \{(d_i, s_i^*, r_i^*)\}_{i=1}^n$, where s_i^* and r_i^* represent the ground-truth score and rationale for dimension d_i .

For dimension d_i with m_i samples, we compute mean squared error and mean absolute error between human scores $\{s_{i,j}^*\}_{j=1}^{m_i}$ and predictions $\{\hat{s}_{i,j}\}_{j=1}^{m_i}$.

$$\text{MSE}_i = \frac{1}{m_i} \sum_{j=1}^{m_i} (s_{i,j}^* - \hat{s}_{i,j})^2, \quad \text{MAE}_i = \frac{1}{m_i} \sum_{j=1}^{m_i} |s_{i,j}^* - \hat{s}_{i,j}| \quad (2)$$

Overall metrics average across all n dimensions:

$$\text{AVGMSE} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i, \quad \text{AVGMAE} = \frac{1}{n} \sum_{i=1}^n \text{MAE}_i \quad (3)$$

Additionally, we evaluate reasoning quality by measuring semantic alignment between generated rationales $\{\hat{r}_{i,j}\}$ and reference rationales $\{r_{i,j}^*\}$ via LLM-based assessment. For each dimension d_i , we compute the average reason quality score:

$$\text{RQ}_i = \frac{1}{m_i} \sum_{j=1}^{m_i} \text{Consistency}(\hat{r}_{i,j}, r_{i,j}^*) \quad (4)$$

where $\text{Consistency}(\cdot, \cdot) \in [0, 1]$ measures rationale consistency. The overall reason quality score averages across all dimensions:

$$\text{RQS} = \frac{1}{n} \sum_{i=1}^n \text{RQ}_i \quad (5)$$

Human-aligned Score (HAS). We define a comprehensive metric, termed the **Human-aligned Score (HAS)**, which integrates score alignment and reasoning quality. First, we normalize alignment metrics to $[0, 1]$ scale:

$$A_{\text{MSE}} = 1 - \frac{\text{AVGMSE}}{16}, \quad A_{\text{MAE}} = 1 - \frac{\text{AVGMAE}}{4} \quad (6)$$

Then compute the weighted alignment and final HAS:

$$\text{Align} = \alpha_1 \cdot A_{\text{MSE}} + \alpha_2 \cdot A_{\text{MAE}} \quad (7)$$

$$\text{HAS} = \beta_1 \cdot \text{Align} + \beta_2 \cdot \text{RQS} \quad (8)$$

SurveyReview Evaluation. In our evaluation protocol, each evaluator is required to output a structured four-dimensional assessment for a given survey paper, consisting of (i) a numeric score and (ii) a supporting rationale for each quality dimension: **Readability**, **Criticalness**, **Comprehensiveness**, and **Structure**.

Readability reflects clarity and ease of understanding; **Criticalness** captures analytical depth and synthesized insight beyond summarization; **Comprehensiveness** evaluates the breadth and relevance of literature coverage; and **Structure** measures the logical organization and coherence of the survey.

Score alignment is measured by AVGMSE and AVGMAE, while rationale quality is quantified by the Reason Quality Score (RQS). The final Human-aligned Score (HAS) integrates both components into a unified metric that quantifies overall alignment with expert peer review.

4 Dataset Construction

Evaluating automatic survey evaluators requires ground-truth data consisting of paired survey papers and expert assessments, denoted as $(\mathcal{P}, \mathcal{E}^*)$, where $\mathcal{P}$ is a survey paper and $\mathcal{E}^*$ represents the ground-truth four-dimensional scores and supporting rationales provided by human reviewers. Such data serves two key purposes: supervised training and alignment-based benchmarking.

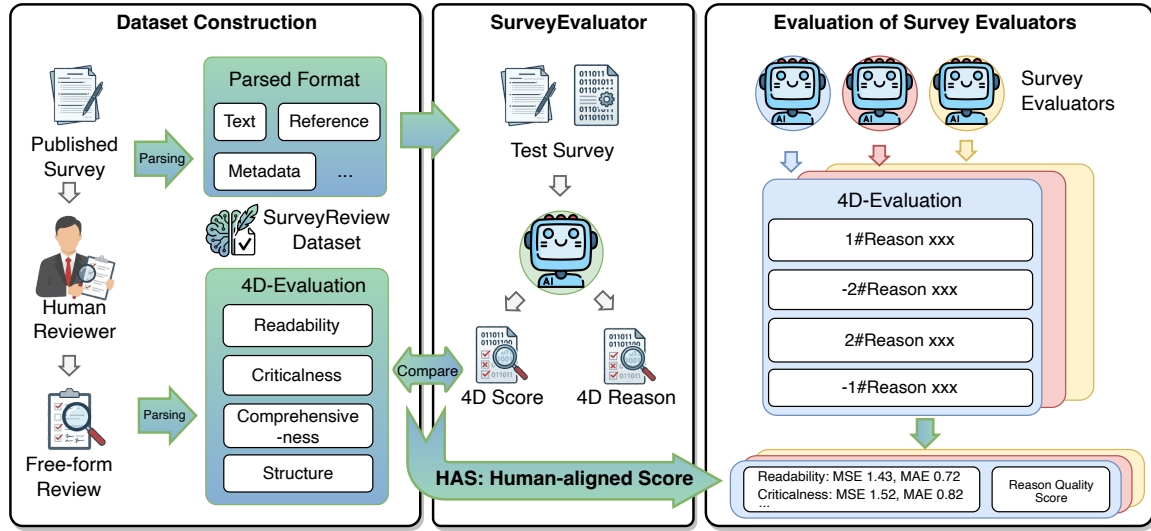


Figure 2: Overview of SurveyReview. Left: dataset construction. Middle: the workflow of a single survey evaluator producing dimension-specific score–rationale pairs. Right: evaluation of survey evaluators under the unified benchmarking protocol.

However, collecting high-quality annotations from domain experts at scale is prohibitively expensive [6]. While existing peer review processes provide expert evaluations of survey papers, these resources are not directly suitable for alignment-based benchmarking. To our knowledge, no existing dataset systematically aggregates survey–review pairs across domains with standardized, dimension-level labels. Moreover, raw review texts are typically unstructured free-form comments, lacking explicit scoring criteria and rationale annotations.

To address these challenges, we construct SURVEYREVIEW through a three-stage pipeline. We **collect** survey papers with peer reviews from multiple academic sources (Section 4.1), **process** each PDF to extract full text and citation structure (Section 4.2), and **annotate** reviews into structured four-dimensional scores and rationales (Section 4.3). The resulting dataset contains 675 surveys with multi-dimensional expert annotations, enabling both supervised training and rigorous evaluation. The overall dataset construction pipeline is illustrated on the left side of Figure 2.

4.1 Data Collection

Data sources. We collect survey papers and their accompanying peer review reports from three publicly available platforms: MO-PRD [12], a large-scale multidisciplinary dataset providing structured open peer review records across the full review lifecycle; F1000Research, an open publishing platform adopting a publish-first, post-publication peer review model with fully transparent reviewer reports and evaluation statuses [23]; and OpenReview, a widely used transparent peer review and conference management system in the machine learning and AI community. Together, these sources offer diverse domains, review processes, and transparency levels, enabling comprehensive and heterogeneous evaluation settings [20].

Survey identification. For each source, we adopt a three-stage filtering strategy to identify survey papers, consisting of keyword-based matching on titles (e.g., “survey” and “review”) to obtain high-recall candidates [31], large language model-based screening of titles and abstracts to verify survey-style content [8], and manual verification to remove ambiguous or misclassified cases, thereby ensuring high precision and data quality. We retain only papers that pass all stages and include substantive peer review reports with explicit quality assessments or critical feedback.

4.2 Survey Paper Processing

After identifying survey papers with accompanying reviews, we retrieve the corresponding PDF files for downstream processing and analysis.

PDF parsing. We extract structured information from the PDFs using two complementary tools. Marker [14] converts PDFs into clean markdown text, providing full paper content for evaluator input. Grobid [1] parses PDFs into structured XML, extracting reference lists, citation contexts, and bibliographic metadata. This dual-tool approach yields comprehensive paper representations, including full text, citation graphs, and metadata.

Extracted information. The processing pipeline produces:

- Full-text content (via Marker): complete paper text for evaluation
- References (via Grobid): reference lists, citation contexts, and locations
- Metadata: title, authors, abstract, and publication year

This structured representation supports both evaluator training (using full text) and knowledge augmentation (using citation analysis).

Datasets	Fields	#Surveys	Metadata	Real Reviews	Balanced	Fine-grained Annotation
SurveyScope	CS	46	✓	✗	✗	✗
SurveyBench	CS	100	✓	✗	✗	✗
SurGE	CS	205	✓	✗	✗	✗
DeepSurvey-Bench	Mixed	163	✓	✗	✗	✗
SurveyReview	Mixed	675	✓	✓	✓	✓

Table 1: Comparison of survey datasets. Real Reviews denotes the use of authentic peer review reports. Balanced indicates availability of both high- and low-quality surveys. Fine-grained Annotation denotes structured dimension-level scores with rationales.

4.3 Annotation Methodology

We convert free-form peer review reports into structured dimension-level annotations. For each survey–review pair, annotators assign scores and extract supporting rationales for four predefined dimensions: *Readability*, *Structure*, *Comprehensiveness*, and *Criticalness*.

The resulting ground-truth expert evaluation for each survey paper is denoted as $\mathcal{E}^* = \{(d_i, s_i^*, r_i^*)\}_{i=1}^n$, where s_i^* and r_i^* represent the annotated score and corresponding rationale for dimension d_i .

Annotation guidelines specify task definition, dimension criteria, rationale extraction rules, and scoring standards. Detailed definitions, examples, and platform descriptions are provided in Appendix A.

All samples are annotated by trained annotators. To assess annotation reliability, a subset of the dataset is independently annotated by multiple annotators. Inter-annotator agreement on this subset, measured by Krippendorff’s alpha, is 0.74.

4.4 Dataset Statistics

The SurveyReview dataset comprises 675 survey papers and 1,630 associated peer review reports with structured four-dimensional annotations.

As summarized in Table 1, SurveyReview differs from existing survey benchmarks in three key aspects. First, it is constructed from authentic peer review reports rather than synthetic judgments or reference-only comparisons. Second, the dataset contains both positive and negative quality signals, reflecting the naturally critical distribution of real-world reviews. Third, each sample is annotated along four predefined quality dimensions with explicit supporting rationales, enabling fine-grained supervision and evaluation.

Source composition. Reviews are collected from F1000Research (1,058; 64.9%), MOPRD (375; 23.0%), and OpenReview (197; 12.1%), providing heterogeneous domains and review styles.

Score distribution. Dimension-wise statistics are reported in Table 5. The overall score distribution is: -2 (4.46%), -1 (66.25%), $+1$ (24.73%), and $+2$ (4.56%). The distribution is skewed toward negative assessments, with extreme scores (± 2) accounting for a small proportion of annotations.

Train–test split. The dataset is split into 1,216 training samples (75%) and 414 test samples (25%), stratified by data source to preserve domain proportions. The split is performed at the paper level: all reviews associated with the same survey paper are assigned to

the same partition, ensuring that no paper appears in both training and test sets.

5 Methods

Under the task formulation in Section 3, survey evaluation aims to learn a mapping $f_\theta : \mathcal{P} \rightarrow \hat{\mathcal{E}}$, where

$$\hat{\mathcal{E}} = f_\theta(\mathcal{P}) = \{(d_i, \hat{s}_i, \hat{r}_i)\}_{i=1}^n$$

denotes the predicted dimension-specific score–rationale pairs for a survey paper $\mathcal{P}$.

To realize this mapping in practice, we develop **SurveyAlign**, a modular evaluator that implements f_θ under the unified evaluation protocol. SurveyAlign combines supervised fine-tuning (SFT) for surface-level dimensions (*Readability* and *Structure*) with dimension-specific Knowledge Augmentation (KA) for knowledge-intensive dimensions (*Comprehensiveness* and *Criticalness*), and further applies inference-time consensus aggregation to improve prediction stability.

5.1 Supervised Fine-Tuning (SFT)

In supervised fine-tuning, we train an instruction-following LLM on SurveyReview annotations to learn reviewer-aligned scoring and rationale generation. We construct instruction-tuning examples consisting of instruction, input, and output fields, each corresponding to a single evaluation dimension. The input contains the survey text, and the output includes a numeric score paired with a human-written rationale. We adopt the same dimension-specific prompting scheme as in Section ??, ensuring consistency between prompt-based evaluation and supervised training.

We build dimension-specific training sets using annotated reviews. Qwen3-32B [27] is fine-tuned with LoRA [7] for parameter-efficient adaptation, learning the mapping from survey text to dimension-specific score–rationale pairs.

Implementation details are provided in Appendix D.

5.2 Knowledge-Augmented Training

The four evaluation dimensions exhibit different dependency structures. *Readability* and *Structure* primarily rely on text-internal signals such as fluency and organization. In contrast, *Comprehensiveness* and *Criticalness* require reasoning beyond the survey text itself. *Comprehensiveness* involves assessing literature coverage relative to the surrounding research landscape, while *Criticalness* requires analytical synthesis across multiple works. These dimensions therefore benefit from explicit Knowledge Augmentation (KA).

5.2.1 Reference Expansion. Given a survey paper $\mathcal{P}$, we extract its reference set R . Using a citation graph constructed from AMiner [21], we expand R to obtain an augmented citation neighborhood R^+ by retrieving papers that have citation relationships with items in R . This expansion provides a structured view of the local research landscape surrounding the survey.

5.2.2 Dimension-Specific Knowledge Augmentation. Since R^+ can exceed the available context window, we construct compact, dimension-specific KA representations before injecting external knowledge into the model input.

Comprehensiveness-specific Knowledge Augmentation (Comp-KA). Comprehensiveness focuses on coverage completeness. We therefore identify structurally important works in R^+ based on their connectivity to the original reference set R . Candidate papers are ranked by citation links with R , and the top-ranked papers are selected together with the original references. The resulting Comp-KA representation is serialized as structured metadata (title, year, citation count), highlighting representative and influential works in the citation neighborhood. This allows the model to reason about potential coverage gaps.

Criticalness-specific Knowledge Augmentation (Crit-KA). Criticalness requires assessing analytical depth and synthesized insight. Unlike coverage evaluation, this dimension depends on modeling the evolution and interaction of ideas over time. We therefore construct a temporally structured representation from R^+ . Papers are filtered using adaptive thresholds derived from reference statistics (e.g., citation count and publication year), forming a representative candidate set. Candidates are sorted chronologically and grouped under a token budget, each block summarized, and the summaries further compressed into a fixed-length representation (Crit-KA). This preserves coarse temporal evolution signals to support cross-paper analytical reasoning.

Detailed construction procedures are provided in Appendix F.

5.2.3 Training with Knowledge-Augmented Inputs. For knowledge-intensive dimensions, the corresponding KA representation is prepended to the survey text as part of the model input.

For *Comprehensiveness*, supervised fine-tuning (SFT) is performed on inputs of the form [Comp-KA; $\mathcal{P}$].

For *Criticalness*, we adopt a two-stage strategy. First, SFT is conducted on inputs augmented with Crit-KA. We then apply Direct Preference Optimization (DPO) [15] under the same augmented setting. Criticalness judgments are inherently more subjective and require nuanced synthesis beyond factual grounding. DPO further aligns the model toward human-preferred analytical reasoning.

Separate LoRA-adapted models [7] are trained for each dimension, enabling dimension-specific supervision and optimization strategies.

5.3 Model Composition and Inference

Test-Time Scaling via Consensus Aggregation. To reduce stochastic variance in generative scoring, we apply a lightweight test-time aggregation strategy inspired by prior work on scaling test-time

compute for LLMs [2, 26]. For each evaluation dimension, the corresponding model is executed multiple times with sampling, producing independent score–rationale pairs. The final score is determined via majority voting across runs, and the rationales are consolidated into a unified explanation consistent with the consensus score. This strategy introduces no additional training cost and improves prediction stability purely through inference-time aggregation. Detailed configurations are provided in Appendix E.

Overall Construction. SurveyAlign is designed as a modular evaluation framework composed of four dimension-specialized sub-models: *SurveyAlign-Read*, *SurveyAlign-Crit*, *SurveyAlign-Comp*, and *SurveyAlign-Stru*. Each sub-model is independently fine-tuned with LoRA to focus on a single evaluation dimension, allowing dimension-specific supervision signals and optimization strategies.

For the surface-level dimensions (*Readability* and *Structure*), we adopt supervised fine-tuning (SFT) (Section 5.1) to improve score calibration and rationale consistency. For the knowledge-intensive dimensions (*Criticalness* and *Comprehensiveness*), we further incorporate knowledge-augmented training (Section 5.2) to enhance evidence grounding and coverage assessment.

At inference time, each dimension-specialized sub-model applies the consensus-based test-time scaling strategy described above, and the resulting dimension-level score–rationale pairs are combined into a unified multi-dimensional evaluation output.

6 Experiments And Analysis

6.1 Experimental Setup

Evaluation Metrics. Following the task formulation in Section 3, we evaluate automatic survey evaluators along two aspects: *score alignment* and *rationale quality*.

For each evaluation dimension, alignment with human reviewers is measured using mean squared error (MSE) and mean absolute error (MAE) between predicted scores and annotated ground-truth scores. Lower values indicate better alignment.

To assess rationale quality, we compute the Reason Quality Score (RQS), which measures semantic consistency between generated rationales and human-written explanations under a standardized evaluation prompt (Appendix G). Higher RQS indicates better reasoning fidelity.

We further aggregate alignment and rationale quality into a unified **Human-aligned Score (HAS)**. Unless otherwise specified, we set $\alpha_1 = 0.4$, $\alpha_2 = 0.6$ for combining MSE and MAE into an alignment score, and $\beta_1 = 0.75$, $\beta_2 = 0.25$ for combining alignment and RQS into HAS. Lower MSE/MAE and higher RQS/HAS indicate better performance.

Implementation Details. Our proposed evaluator, **SurveyAlign**, is built upon Qwen3-32B [27]. We apply Low-Rank Adaptation (LoRA) [7] for parameter-efficient supervised fine-tuning. Dimension-specific models are trained following the framework described in Section 5. Detailed hyperparameters and training configurations are provided in Appendix D.

Input Format. All evaluated methods operate under a unified prompting protocol (Appendix C). For zero-shot LLM-as-a-Judge baselines and our fine-tuned models, we instantiate a standardized

Table 2: Main results across all evaluated methods. AVG: average across the four evaluation dimensions. RQS: Reason Quality Score. HAS: Human-aligned Score

Method	Model	Read.		Crit.		Comp.		Stru.		AVG		RQS ↑	HAS ↑
		MSE ↓	MAE ↓	MSE ↓	MAE ↓	MSE ↓	MAE ↓	MSE ↓	MAE ↓	MSE ↓	MAE ↓		
SurveyAlign	Qwen3-32B-lora	1.43	0.72	1.52	0.82	1.26	0.56	1.29	0.65	1.38	0.69	0.36	0.74
<i>Zero-shot</i>													
Prompt	Qwen3-32B	3.05	1.45	3.24	1.51	3.22	1.54	3.35	1.53	3.21	1.51	0.36	0.61
Prompt	GPT-5.2	2.13	1.07	1.97	0.97	2.04	1.08	2.98	1.47	2.28	1.15	0.42	0.68
Prompt	Gemini-3-pro	3.84	1.52	2.25	1.00	3.91	1.49	5.76	2.11	3.94	1.53	0.29	0.58
Prompt	DeepSeek-v3.2	4.78	1.88	2.49	1.15	4.59	1.82	4.02	1.76	3.97	1.65	0.37	0.58
Prompt	GLM-4.7	3.43	1.50	2.58	1.21	3.66	1.57	4.83	1.95	3.62	1.56	0.37	0.60
Prompt	Claude-Opus-4.5	2.91	1.29	1.88	0.88	2.66	1.23	3.65	1.58	2.77	1.25	0.48	0.68
<i>Prior Work</i>													
LLMMapReduce	gemini-3-pro	2.95	1.27	3.04	1.29	5.04	1.83	6.13	2.24	4.29	1.66	–	–
DR-Bench	GPT-5.2	3.06	1.39	2.51	1.13	2.75	1.29	–	–	–	–	–	–
SurveyX	GPT-5.2	–	–	2.14	1.00	–	–	3.19	1.43	–	–	–	–

prompt template consisting of (1) the evaluation dimension definition, (2) scoring rubric, (3) explicit output format specification, and (4) the full survey text.

For knowledge-augmented settings, the corresponding dimension-specific Knowledge Augmentation (KA) representation is prepended to the survey text without modifying the remaining prompt structure.

For prior evaluators, we follow their original prompting strategies and pipeline designs while adapting their outputs to the standardized score–rationale format required by our benchmark.

Baselines. We compare SurveyAlign against two categories of baselines:

(i) **Prompt-based LLM-as-a-Judge.** Off-the-shelf instruction-tuned LLMs are used as zero-shot evaluators without task-specific training on SurveyReview. For each survey paper, models are executed independently for each evaluation dimension, conditioned on the corresponding dimension-specific rubric. We evaluate GPT-5.2 [18], Claude-Opus-4.5-Thinking, Gemini-3-Pro-Preview, DeepSeek-v3.2 [4], GLM-4.7 [22], and Qwen3-32B [27] in their zero-shot settings.

(ii) **Prior Survey Evaluators.** We additionally compare with LLMMapReduce [24], DR-Bench [5], and SurveyX [10]. These systems are evaluated using their original methodologies, with outputs converted to the standardized score–rationale format for fair comparison under our benchmark.

6.2 Main Results

Overall alignment. Table 2 reports the alignment performance of all evaluators on SurveyReview. Our fine-tuned evaluator, **SurveyAlign** (Qwen3-32B-LoRA), achieves the best performance across all four dimensions. It obtains the lowest MSE and MAE on every metric and achieves the highest overall HAS score of 0.74, substantially outperforming all zero-shot baselines and prior work.

Comparison with zero-shot LLM judges. Even the strongest zero-shot model (GPT-5.2) reaches a HAS of 0.68, whereas SurveyAlign achieves 0.74. Beyond the overall score, SurveyAlign consistently

attains lower MSE and MAE on every dimension, with regression errors reduced by up to 4×. This consistent margin indicates that task-specific supervision is necessary to approach reviewer-level consistency.

Dimension-wise improvements. Performance gains are observed across both surface-level and knowledge-intensive dimensions. Readability and Structure, which rely primarily on text-internal signals, benefit substantially from supervised fine-tuning: Readability MSE decreases from 2.13 (GPT-5.2 zero-shot) to 1.43, and Structure from 2.98 to 1.29. The particularly large reduction on Structure suggests that document-level organization requires task-specific calibration.

For the knowledge-intensive dimensions—Criticalness and Comprehensiveness—incorporating structured external knowledge further improves alignment. Criticalness MSE decreases to 1.52 and Comprehensiveness to 1.26, surpassing all zero-shot baselines. These results highlight the importance of explicit knowledge modeling for coverage reasoning and analytical depth.

Rationale quality. Beyond numeric scores, we evaluate reasoning quality using RQS. While Claude-Opus-4.5-Thinking achieves the highest zero-shot RQS (0.48), it underperforms in score alignment. In contrast, SurveyAlign maintains competitive rationale quality (0.36) while delivering substantially better quantitative alignment, demonstrating a favorable trade-off between explanation quality and scoring accuracy.

Comparison with prior evaluators. Prior systems such as DR-Bench, LLMMapReduce, and SurveyX rely primarily on prompting designs rather than data-driven supervision, resulting in weaker alignment or incomplete dimensional coverage. In contrast, SurveyAlign is trained directly on human-annotated data and achieves strong, consistent performance across all four dimensions.

Takeaway. Overall, results demonstrate that supervised alignment is crucial for reliable survey evaluation, and that knowledge augmentation further strengthens performance on knowledge-intensive criteria.

6.3 Ablation Study of SurveyAlign

Table 3: Ablation of the SurveyAlign-Comp submodel on the Comprehensiveness (Comp) dimension.

Setting (SurveyAlign-Comp)	MSE ↓	MAE ↓
SFT	1.36	0.60
SFT + Comp-KA	1.26	0.56

Table 4: Ablation of the SurveyAlign-Crit submodel on the Criticalness (Crit.) dimension.

Setting (SurveyAlign-Crit)	MSE ↓	MAE ↓
SFT	1.96	0.93
SFT + Crit-KA	2.04	1.02
SFT + DPO	1.90	0.99
SFT + Crit-KA + DPO	1.52	0.82

6.3.1 Comprehensiveness. **Comprehensiveness-specific Knowledge Augmentation (Comp-KA)** is evaluated on the Comprehensiveness dimension under identical training settings. on the Comprehensiveness dimension under identical training settings. As shown in Table 3, incorporating Comp-KA reduces MSE from 1.36 to 1.26 and MAE from 0.60 to 0.56. The consistent improvement across both metrics indicates that structured bibliographic context improves modeling of survey coverage and completeness.

6.3.2 Criticalness. We further examine the interaction between **Criticalness-specific Knowledge Augmentation (Crit-KA)** and Direct Preference Optimization (DPO) on the Criticalness dimension (Table 4). Crit-KA alone does not improve performance and slightly increases error, while DPO alone yields only marginal gains. In contrast, combining Crit-KA with DPO substantially reduces MSE (1.52) and MAE (0.82), achieving the best performance. These results suggest that Criticalness evaluation requires both access to relevant external knowledge and alignment of reasoning behavior, highlighting the complementary roles of dimension-specific Knowledge Augmentation and preference optimization.

7 Conclusion

We presented SurveyReview, a reviewer-aligned, multi-dimensional benchmark for evaluating survey papers. By structuring peer-review reports into four quality dimensions with rationales, the framework enables quantitative measurement of alignment between automatic evaluators and human reviewers.

Experiments show that task-specific supervision substantially improves alignment compared to zero-shot judging, especially on subjective dimensions such as Criticalness and Structure. Dimension-wise and ablation analyses further reveal distinct roles of knowledge augmentation, preference optimization, and inference strategies.

The benchmark provides standardized splits, evaluation metrics, and diagnostic signals for studying reviewer-aligned assessment.

We hope this benchmark facilitates more transparent and measurable progress in automated survey evaluation.

8 Ethical Considerations

This work adheres to the ACL Code of Ethics. The SurveyReview benchmark is built exclusively from publicly available survey papers and open peer review materials originating from OpenReview, F1000Research, and MOPRD. We do not claim ownership of any third-party texts. We follow the licensing terms, terms of service, and access controls of each source. Consistent with these requirements, we will maximize redistribution of materials that are explicitly redistributable, and for materials that are not redistributable we will provide stable identifiers, provenance, and retrieval instructions so that users can obtain them directly from the original sources under the applicable terms. For metadata and structured annotations that are legally redistributable, we will make them publicly available to the greatest extent permitted, accompanied by clear documentation to facilitate reproducibility and further research.

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A Annotation Details

This appendix provides detailed guidelines for the annotation protocol used to construct the SURVEYREVIEW dataset.

A.1 Task Definition

Annotators are instructed to read the peer review report of a survey paper and, for each quality dimension, extract:

- (1) A discrete score from $\{-3, -2, -1, 0, +1, +2\}$ reflecting the reviewer’s sentiment toward that dimension.

Scores of 0 (not mentioned) and -3 (unjudgeable or ambiguous sentiment) are treated as special cases and are excluded from the final supervised training and evaluation set.

- (2) The most relevant supporting sentence(s) from the review text that justify the assigned score (rationale).

If a dimension is mentioned but the sentiment is ambiguous or cannot be clearly determined, annotators may assign -3 (unjudgeable), and such cases are excluded from supervised learning.

A.2 Dimension Definitions

Comprehensiveness. Evaluates the completeness and relevance of the manuscript’s references. Annotators assess whether key works are cited, important literature is omitted, or outdated/irrelevant sources are used.

Criticalness. Evaluates the originality and depth of the survey, focusing on whether it provides new insights, critical perspectives, or meaningful synthesis beyond summarizing prior work. Related keywords: *insight, novel perspective, depth*.

Readability. Evaluates the overall clarity and presentation quality of the manuscript, including whether the writing is clear, well-explained, easy to follow, or difficult to understand. Related keywords: *writing, presentation, easy to follow, hard to follow*.

Structure. Evaluates the logical organization and coherence of sections and subsections, including whether ideas are well-structured and transitions between parts are smooth and logical.

A.3 Scoring Criteria

Scores are defined as follows:

-2 (*Strong Negative*). Strongly negative evaluation, typically appearing in the weaknesses section with explicit critical language (e.g., “extremely limited,” “serious flaw”). The tone is clearly critical without balancing positive remarks.

-1 (*Negative*). Moderately negative evaluation, often introduced by contrastive expressions such as “however” or “although,” highlighting limitations or areas for improvement.

0 (*Not Mentioned*). The review does not explicitly address this dimension. No rationale is required.

$+1$ (*Positive*). Moderately positive evaluation, usually appearing in the strengths section with brief or mild praise.

$+2$ (*Strong Positive*). Strongly positive evaluation, characterized by emphatic language (e.g., “extremely well-written,” “excellent and comprehensive”) or multiple explicit commendations.

-3 (*Unjudgeable*). The dimension is mentioned, but the sentiment is mixed, vague, or internally inconsistent, making it impossible to assign a clear polarity. Such cases are excluded from training.

A.4 Rationale Guidelines

For each dimension, annotators extract the sentence(s) from the review that most directly support the assigned score.

- Rationales should be as complete as possible; partial fragments should be expanded when necessary to preserve meaning.
- If multiple supporting statements are present, annotators may list them as separate points (1), (2), (3), etc.
- If a dimension is scored as 0, no rationale is required.

The extracted rationales serve as explicit supervision signals for interpretable and evidence-grounded evaluation.

A.5 Annotation Examples

This subsection presents representative review excerpts for each quality dimension and score level.

A.5.1 Comprehensiveness.

Score -2.

- “There are major flaws in the article.”
- “There are many survey articles on WSN; authors must compare their surveys with existing surveys.”
- “Authors are encouraged to cite the latest papers from 2023.”

Score -1. “However, the authors have not explained why they chose Google Scholar as the sole database for their search.”

Score 0. No explicit comment on literature coverage.

Score +1. “This paper systematically examined all published literature (481 papers).”

Score +2.

- “The article very usefully includes current resources and guidelines.”
- “The information is very comprehensive and relevant to the aims of the article.”

A.5.2 Criticalness.

Score -2. “The authors suggest future analysis without specifying how this analysis will be conducted.”

Score -1. “While the article comments on issues and challenges, expanding this discussion could enhance the practical relevance.”

Score 0. No explicit comment on originality or analytical depth.

Score +1. “The authors have raised an important point about differences in cut-offs.”

Score +2. “This is a very interesting and useful article, which makes a contribution to the knowledge...”

A.5.3 Readability.

Score -2. “The section ‘Issues and challenges..’ needs more explanation and justification.”

Score -1. “The method ‘one-term in one-source’ is not well known to the research community.”

Score 0. No explicit comment on clarity or presentation.

Score +1. “The language is quite clear and well-written.”



Figure 3: Annotation Platform

Score +2.

- “The introduction is right on target and reads well.”
- “The discussion also reads well and adjusts well to the findings.”

A.5.4 Structure.

Score -2.

- “Discussion is very unstructured.”
- “There is a mismatch in the databases in the PRISMA Flow diagram.”
- “There is no mention if case reports, letters and editorials were included or not.”

Score -1. “The studies are discussed as isolated ideas that do not inform or complement each other.”

Score 0. No explicit comment on organization or coherence.

Score +1. “The article is logically organized and uses relevant literature.”

Score +2.

- “The data paper describes in detail not only the dataset but also the annotation and validation processes.”
- “The document is excellently suited for teaching.”

A.6 Annotation Platform

Figure 3 shows the annotation platform.

B Dataset Details

This appendix provides detailed statistics of the SURVEYREVIEW dataset, including dimension-wise score distributions and rationale density.

B.1 Dimension-Level Statistics

Table 5 reports summary statistics for each quality dimension: Readability, Structure, Comprehensiveness, and Criticalness.

Table 5: Dimension-wise statistics showing score distributions and rationale density.

Dimension	Mean Score	Std	Avg. R
Readability	−0.36	0.94	3.54
Structure	−0.40	0.83	3.07
Comprehensiveness	−0.29	0.90	2.84
Criticalness	−0.17	0.98	2.80

Score statistics. Mean scores range from −0.40 (Structure) to −0.17 (Criticalness), indicating a moderate skew toward negative evaluations across dimensions. Standard deviations are close to 1.0 for all dimensions, suggesting substantial variation in reviewer judgments.

Rationale density. The column “Avg. R” reports the average number of rationale snippets per annotated instance for each dimension. Each snippet corresponds to a distinct comment segment extracted from the original peer review report. On average, reviewers provide between 2.80 and 3.54 rationale segments per dimension, reflecting fine-grained feedback.

Annotation granularity. Each survey paper may receive multiple review reports. Dimension-level scores and rationales are extracted independently for each review report, and statistics in Table 5 are computed at the (paper, review, dimension) level.

C Evaluation Prompting Protocol

C.1 Unified Prompt Template

All evaluation prompts follow a unified four-component structure:

- (1) A role instruction defining the evaluator as an academic reviewer and introducing the content to be assessed
- (2) Definition of the target evaluation dimension
- (3) Description of the scoring scale
- (4) Explicit instructions to produce a numeric score with a supporting rationale

Format demonstrations are included to encourage consistent responses.

Please act as a research paper reviewer and evaluate the following survey from the perspective of [Target Dimension].

[Knowledge Augmentation Description—used only in KA setting]

[Definition of Target Dimension]

[Scoring Scale Description]

Output Requirements:

Provide:

- (1) A brief reasoning explaining your evaluation.
- (2) A numeric score chosen from {−2, −1, 1, 2}.

Output Format:

Reason\$\$\$X\$\$\$

Example:

[Example rationale]\$\$\$[Example Score]\$\$\$

{Knowledge Augmentation Content – used only in KA setting}
{FullSurveyText}

C.2 Dimension Definitions and Scoring Scales

All dimensions are scored on a four-point scale −2, −1, 1, 2, where higher values indicate stronger alignment with reviewer expectations.

C.2.1 Readability.

Definition:

Readability evaluates the overall presentation quality, focusing on clarity, coherence, and ease of understanding.

Scoring Scale:

- −2 = Very hard to understand; poor clarity or structure
- −1 = Somewhat unclear; ideas or organization need improvement
- 1 = Basically clear; statements are understandable
- 2 = Very clear and well-presented throughout

Example (Score = 2):

The writing is fluent and well-organized, with a clear logical flow across sections.

C.2.2 Criticalness.

Definition:

Criticalness assesses the manuscript's originality and analytical depth. It examines whether the survey provides new insights, critical perspectives, or meaningful contributions beyond summarizing prior work.

Scoring Scale:

- −2 = Lacks originality; no meaningful new insights
- −1 = Limited novelty; mostly reiterates existing knowledge
- 1 = Reasonable originality; offers some new perspectives
- 2 = Highly original and insightful; strong critical depth

Example (Score = 2):

The survey synthesizes prior work into a novel framework and identifies key unresolved research gaps.

C.2.3 Comprehensiveness.

Definition:

Comprehensiveness measures the completeness and relevance of literature coverage, including whether key works are cited and irrelevant or outdated sources are avoided.

Scoring Scale:

- −2 = Major gaps; important works missing
- −1 = Some relevant works missing or outdated sources used
- 1 = Adequate coverage with minor gaps
- 2 = Comprehensive and highly relevant coverage

Example (Score = 2):

The references cover all major research directions and include up-to-date representative works.

C.2.4 Structure.

Definition:

Structure evaluates the logical organization of sections and subsections, ensuring coherent progression of ideas and clear connections between components.

Scoring Scale:

-2 = Poor organization; lacks logical flow
 -1 = Some structural issues; connections unclear
 1 = Reasonably organized; mostly logical order
 2 = Excellently structured; seamless logical flow

Example (Score = 2):

The survey follows a clear progression, with well-defined sections and consistent transitions.

C.3 Shared Prompt Template for Zero-Shot and SFT

Please act as a research paper reviewer and evaluate the following survey from the perspective of [Target Dimension].

[Definition of Target Dimension]

[Scoring Scale Description]

Output Requirements:

Provide:

(1) A brief reasoning explaining your evaluation.
 (2) A numeric score chosen from {-2, -1, 1, 2}.

Output Format:

Reason\$\$\$X\$\$\$

Example:

[Example rationale]\$\$\$[Example Score]\$\$\$

Survey to Evaluate:

{FullSurveyText}

C.4 Knowledge Augmentation Prompt

In the Knowledge Augmentation (KA) setting, dimension-specific external reference information is provided as supplementary context. While the overall prompt structure remains consistent with the unified template, the content of the augmentation block differs across dimensions.

C.4.1 General KA Framework.

Please act as a research paper reviewer and evaluate the following survey from the perspective of [Target Dimension].

[Knowledge Augmentation Description]

[Definition of Target Dimension]

[Scoring Scale Description]

Output Requirements:

Provide:

(1) A brief reasoning explaining your evaluation.
 (2) A numeric score chosen from {-2, -1, 1, 2}.

Output Format:

Reason\$\$\$X\$\$\$

Example:

[Example rationale]\$\$\$[Example Score]\$\$\$

{KA_Content}

{FullSurveyText}

C.4.2 KA: Criticalness. For the Criticalness dimension, the [Knowledge Augmentation Description] block is instantiated as follows:

You will be provided with two parts of information:

(1) Expanded Work Summary:

A summary of relevant works that are not cited in the original survey but are related to its topic. The summary is organized chronologically and includes time span, research topics, methodological characteristics, and key findings. These works represent potentially important literature that the survey may have missed.

(2) Original Survey:

The full text of the survey paper being evaluated.

C.4.3 KA: Comprehensiveness. For the Comprehensiveness dimension, the [Knowledge Augmentation Description] block is instantiated as follows:

You will be provided with three parts of information:

(1) Original Bibliography:

The list of references cited in the survey paper.

(2) Expanded Reference Set:

An expanded set of references retrieved based on the survey topic. These references are provided for comparison and contextual grounding.

(3) Original Survey:

The full content of the survey paper.

The expanded reference set should be used as context to assess whether the original bibliography is comprehensive and relevant.

D SFT Details

We perform supervised fine-tuning (SFT) based on Qwen3-32B using parameter-efficient Low-Rank Adaptation (LoRA). Each evaluation dimension is trained as a separate model.

LoRA configuration. We adopt LoRA with rank $r = 8$ and scaling factor $\alpha = 16$, without dropout (0.0). LoRA is applied to all attention and feed-forward projection layers, including q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj. Only LoRA parameters are updated, while the base model weights remain frozen.

Training setup. Models are trained for 15 epochs using AdamW with a learning rate of 5×10^{-5} and a cosine learning rate scheduler. We use 20 warmup steps. The per-device batch size is 1 with gradient accumulation steps of 16, resulting in an effective batch size of 16. The maximum sequence length is 16,384 tokens.

Optimization and efficiency. Training is conducted in fp16 precision with gradient checkpointing enabled. We use DeepSpeed ZeRO-3 for memory optimization and FlashAttention-2 for efficient long-context attention.

Data split. We use a 90% / 10% train-validation split and select the best checkpoint based on validation loss.

E Test Time Scaling Details

We implement test-time scaling with $K = 5$ independent sampling runs per dimension, using temperature $T = 0.7$.

Consensus scoring. For a given dimension, let $\{s_i\}_{i=1}^K$ denote the scores generated across runs. The final score is determined via majority voting:

$$s_{\text{final}} = \arg \max_s \sum_{i=1}^K \mathbb{I}[s_i = s]. \quad (9)$$

In the case of ties, we select the score with the highest average log-probability under the generating model.

Rationale synthesis. All K rationales, together with the dimension-specific rubric and the consensus score s_{final} , are provided to GPT-5.2 as a meta-synthesizer. The model is prompted to generate a unified explanation that consolidates recurring evaluation points while remaining consistent with s_{final} .

The meta-synthesis prompt is shown below:

You are a Super Reviewer tasked with synthesizing multiple review outcomes into a single, comprehensive final review.

You will be given:

1. The original review instruction used by individual reviewers
2. The consensus score (mode score) across runs
3. Multiple reasoning explanations from different runs

Your task is to:

- Analyze all provided reasoning explanations
- Identify key themes and recurring points
- Synthesize a coherent final reasoning
- Ensure the explanation justifies the consensus score

- Keep the reasoning comprehensive yet concise

The final score must be the provided consensus score. Output the reasoning in the same format as individual reviews.

This inference-time aggregation improves evaluation stability without requiring additional fine-tuning.

F Knowledge Augmentation Details

This appendix describes the construction of the dimension-specific Knowledge Augmentation (KA) content used for knowledge-intensive dimensions.

F.1 Comprehensiveness-specific Knowledge Augmentation (Comp-KA)

Let R denote the reference set cited by a survey, and R^+ denote the expanded citation neighborhood constructed by the toolkit.

Connectivity scoring. For each candidate paper $p \in R^+$, we define its connectivity to R as:

$$\text{score}(p) = \sum_{r \in R} \mathbb{I}[p \leftrightarrow r],$$

where $(p \leftrightarrow r)$ indicates that p and r have a citation relationship (either direction).

Selection and serialization. Given $n = |R|$, we select the top- $2n$ papers in R^+ ranked by $\text{score}(p)$ and combine them with the original set R :

$$\text{Comp-KA} = R \cup \text{Top}_{2n}(R^+).$$

The resulting Comp-KA content is serialized as a list of paper entries, each containing title, publication year, and citation count, providing structured signals for coverage assessment.

F.2 Criticalness-specific Knowledge Augmentation (Crit-KA)

We construct a temporally structured representation from R^+ to support analysis-oriented evaluation.

Thresholding and candidate set. We compute adaptive thresholds from the original reference set R :

$$\tau_c = \text{Percentile}_\alpha(\{\text{citations}(r)\}_{r \in R}), \quad \tau_y = \text{Percentile}_\beta(\{\text{year}(r)\}_{r \in R}),$$

where α and β are fixed hyperparameters.

We then form a candidate set:

$$C = \{p \in R^+ \mid \text{citations}(p) \geq \tau_c \vee \text{year}(p) \geq \tau_y\}.$$

Intuitively, C includes papers that are either highly cited or relatively recent compared with the survey's references.

Stage 1: Temporal block summarization under a token budget. We sort C by publication year in ascending order. Under a predefined token budget B , we sequentially pack (title, abstract) pairs into a buffer until reaching the budget, forming a temporal block.

For each block, we record its year interval $\Delta t = [y_{\min}, y_{\max}]$ and apply an LLM to produce a concise block summary:

$$(\Delta t_i, \text{Summary}_i) = \text{Summarize}(\text{Block}_i).$$

Stage 2: Global compression to a fixed length. Given the set of timestamped block summaries $\{(\Delta t_i, \text{Summary}_i)\}$, we perform a second-stage compression to obtain a final representation of fixed length L :

$$\text{Crit-KA} = \text{Compress}(\{(\Delta t_i, \text{Summary}_i)\}, L).$$

The output preserves coarse temporal structure through Δt_i while enforcing a global length constraint.

G Reason Quality Score

We compute RQS using GPT-5.2 as a meta-evaluator to assess semantic consistency between predicted rationales and reference rationales. The model assigns a score in 0.0, 0.2, 0.4, 0.6, 0.8, 1.0, where higher values indicate stronger alignment.

The evaluation prompt is shown below.

You are an expert evaluator tasked with assessing the quality of AI-generated review reasoning by comparing it against reference reasoning.

Your Task:

Evaluate how well the predicted reasoning aligns with the reference reasoning for a given research paper review.

Input Information:

You will receive:

1. Dimension
2. Dimension Definition
3. Paper Title
4. Reference Reasoning (true_reasons)
5. Predicted Output (pred_output)

Scoring Scale (output ONLY one score):

1.0 Perfect Alignment

- Core arguments are identical
- All key details are covered
- Logical structure matches
- Only minor wording differences

0.8 High Alignment

- Core arguments consistent
- Most key details covered
- Similar logical flow

0.6 Basic Alignment

- Main points consistent
- Some key details covered
- Minor omissions

0.4 Partial Alignment

- Some overlap
- Significant omissions or inconsistencies

0.2 Minimal Alignment

- Few common points
- Major argument differences

0.0 No Alignment

- Completely different arguments
- No substantial overlap

Output Format:

Return ONLY one number:

0.0, 0.2, 0.4, 0.6, 0.8, or 1.0

H Baseline

For all baseline methods, we adopt the latest flagship models recommended or reported in their respective papers to ensure strong and up-to-date performance.

Since different benchmarks use different scoring schemes, we normalize all outputs into our unified score space $\{-2, -1, 1, 2\}$ for fair comparison. When necessary, scores are linearly mapped to the closest corresponding category in our scale.

H.1 LLM MapReduce V2

We follow the prompt design and described in the original paper, using the latest recommended base model for generation.

H.2 DeepResearch-Bench

DeepResearch-Bench requires reference survey answers for certain evaluation settings. As such gold-standard surveys are not available in our dataset, we adopt the prompt and evaluation framework proposed in DeepResearch-Bench while excluding components that rely on reference survey texts. When evaluating DeepResearch-Bench, we align its insight score with our Criticalness dimension.

H.3 SurveyX

We follow the evaluation and prompting configuration described in the SurveyX paper, using its recommended flagship model for generation and assessment.

I Example: Real Review to Structured 4D Evaluation

Paper Title:

Technological tools for the measurement of sensory characteristics in food: A review

Original Review:

This review explores various technological tools, including e-nose, e-tongue, artificial vision systems, and texture analysis instruments used to analyse and assess food properties like quality, composition, maturity, authenticity, and origin. By standardising these characteristics, they enhance existing food products and create new ones that cater to consumers' sensory preferences. This advancement supports growth in the food sector by delivering satisfying sensory experiences to consumers.

The paper exhibits a well-organised structure, with

coherent headings and subheadings that contribute to the overall clarity of the manuscript. The write-up is satisfactory and presented in good English. The figures and tables provided are clear and comprehensible, aiding in understanding the findings.

However, there are some suggestions as follows:

- * The abstract should be rewritten to include the findings of the study.
- * Colour measurement devices were not discussed sufficiently, so more discussion is needed.

Structured 4D Score-Reason Pairs:

Readability

Score: 1

Reasons:

- The write-up is satisfactory and presented in good English.
- The figures and tables provided are clear and comprehensible, aiding in understanding the findings.

Comprehensiveness

Score: 1

Reasons:

- The paper provides a comprehensive overview of various technological tools used in sensory analysis in the food industry.
- It covers a wide range of devices and techniques, including e-nose, e-tongue, artificial vision systems, texture analyzers, electromyographic analysis, and acoustic analysis.
- This comprehensive coverage is valuable for readers interested in understanding the diversity of tools available for sensory analysis.

Criticalness

Score: -1

Reasons:

- However, it would benefit from a more critical evaluation of the limitations and challenges associated with these tools and a discussion of recent advancements in the field.
- This article does not demonstrate an innovative character.
- There are several reviews already published in this area.
- This manuscript should draw attention to real innovations and aspects that have not yet been addressed in the literature.
- However, the authors should provide a wealth of information on the various tools, it lacks critical evaluation and discussion of their

limitations and challenges.

- For instance, the paper does not discuss the cost, maintenance requirements, and calibration challenges that may arise when implementing these tools in food production settings.
- The manuscript does not discuss recent advancements and developments in the field of sensory analysis technology.
- Given the rapid pace of technological innovation, it would be beneficial to include information on any emerging tools or techniques that have been developed since the paper's publication.

Structure

Score: 1

Reasons:

- The paper exhibits a well-organised structure, with coherent headings and subheadings that contribute to the overall clarity of the manuscript.